AlQavi Hasan

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EDUCATION

University of Sheffield | MEng Chemical Engineering | First Year *Sep 2026 - June 2030*
• Societies: Investment Society
• Debating Society | Politics Society MUN | European Affairs Society | SLUGS (E-Sports)

Nottingham High School *Sep 2016 - Aug 2023*
• A Levels - Maths (A*), Physics (A*), Chemistry (A)
• STEM Prefect | Debating Team | President of Engineering Society

WORK AND EXTRACURRICULAR

Head of Back-End | Imperial Algorithmic Trading Society *Aug 2025 - Aug 2026*
• Designed the system architecture for QT Capital, integrating multi-strategy execution and portfolio monitoring
• Deployed QT Capital trading platform, with a focus on robustness and reliability, with an accessible front-end
• Organised lectures with guest speakers on quantitative topics, with average attendance of over 100 people

Head of Marketing | Imperial College Investment Society *Oct 2024 - Aug 2025*
• Grew society to 1500 members from the previous year's peak of 1200 members through effective marketing
• Received high merits from industry professionals on marketing leading to record-breaking sign-ups for events
• Coordinated logistics for events, ensuring smooth progression - room booking, scheduling, marketing

Analyst | QT Capital Commodities Team *Nov 2024 - May 2025*
• Analysed crude oil and metals price drivers, focusing on supply disruptions, OPEC policy, inventory data
• Built and presented spread models (crack, calendar) to identify relative value opportunities
• Produced weekly trade reports highlighting risk, volatility shifts and metrics for portfolio

RELEVANT PROJECTS

Petral Commodities Dashboard | Python | https://petral.xyz *Nov 2025 - Present*
• Built a crude oil trading desk aggregating market, fundamental and physical data from EIA, CFTC, CME, FRED, NOAA and other public sources, with source and freshness provenance across each dataset
• Developed physical-market analytics spanning WTI/Brent forward curves, crack and calendar spreads, Gulf crude differentials, crude assays and Europe/Asia arbitrage models incorporating tanker freight and financing
• Integrated tanker AIS, chokepoint flows and global supply/demand data to monitor crude movements and balances, alongside a JSON API serving curves, inventories, positioning and physical-market datasets

QT Capital Trading Platform | Python | www.alqavi.xyz/demos/qtc *Aug 2025 - Aug 2026*
• Built a platform with real-time data ingestion, and live execution via Alpaca for 10 teams concurrently
• Robust portfolio tracking and price data storage with Parquet pipelines for efficient storage
• Real time tracking of performance metrics including PnL, Sharpe, Sortino, Calmar and drawdowns
• FastAPI endpoints to display leaderboard and metrics on Next.js dashboard

SSVI Surface | Python | GitHub *Oct 2025 - Dec 2025*
• Fitted an arbitrage-free SSVI implied volatility surface using the SLSQP algorithm with constraints
• Achieved Vega-weighted RMSE within 200bps of market option IVs across maturities and strikes
• Bootstrapped a zero-coupon yield curve from US Treasury instruments for discounting and forward construction
• Optimised functions with NumPy vectorisation to achieve performance up to 300% faster

Competitions
• 3rd out of 2170 in CME Group Futures Trading challenge, with WTI futures *Feb 2026*
• IMC Prosperity 3 Competition - Top 10% in the UK *April 2025*
• Bloomberg Trading Challenge - Earnings surprise strategy *Oct 2024*
• Top 100 RSC UK Chemistry Olympiad, achieving gold and placing top 1% in the country *Jan 2023*

ACTIVITIES AND INTERESTS

Technical Skills: Python, MATLAB, MS Office, Pandas, FastAPI, SciPy, NumPy
Interests: Geopolitics, Debating, History, Photography